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Chapter 3 – Licensing

In earlier chapters you were introduced to the various business models employed by a company involved in open source, and several open source licenses were mentioned. But what makes a software license open source? This chapter provides an answer to this question.

In this chapter you will learn about:

- The tenets of intellectual property, copyright and the intents of an open source license.
- An explanation of what makes a license an open source license
- A brief description of some of the most commonly used open source licenses
- Things to keep in mind while choosing a license for a software

3.1 Intellectual property, copyright and licensing: The big picture

Intellectual Property (IP) are legally protected rights that one has over new ideas or creations. Common types of intellectual property include copyrights, trademarks, patents, industrial design rights and trade secrets.

Copyright gives the creator of the original work exclusive rights in terms of usage, distribution and customization of the work.

Some of the privileges copyright provides to the author of software include:

- The right to produce and sell copies of the work
- The right to create derivative works
- The right to sell, transfer or reassign any of the rights granted by copyright to others

The transfer of rights by the author partly or wholly at his own terms is what we refer to as ***licensing***. The term ***license*** means permission. The copyright holder, or ***licensor***, grants another person, known as the ***licensee***, specific permissions to use the work.

Figure 3.1 depicts this relationship between licensor, licensee and license. It also shows that a license is a subset of the privileges that the Copyright Act grants the licensor. The

licensor permits the licensee to use these privileges as agreed between the two parties in the license.

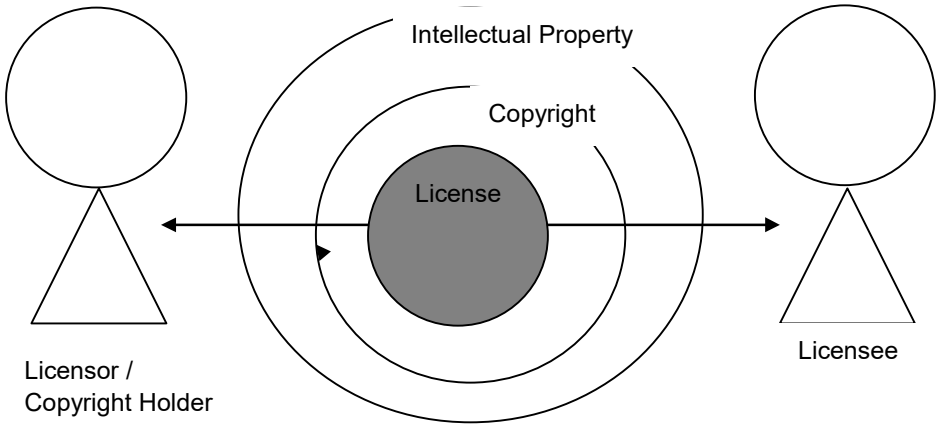


Figure 3.1: Relationship between licensor, licensee and license

3.2 Open source licensing

This section provides a brief history about open source licensing, its intent, and a brief description of commonly used licenses.

3.2.1 History of open source licensing

The open source software movement traces its history to the formation of the Free Software Foundation ("FSF") in 1985 by Richard Stallman. As discussed in *Chapter 1*, Free Software is more of an ideology that emphasizes the freedom users have with the source code and not with the price one pays for the software. In essence, free software is an attempt to guarantee certain rights for both, users and developers. These freedoms include:

- Freedom to execute the program for any reason
- Freedom to examine the source code, see how it works and change it to do what you would like it to do
- Freedom to redistribute the source and keep the money generated from it
- Freedom to modify and redistribute the modified source code

In order to guarantee these freedoms the GNU General Public License (GPL) was created. In short any software licensed under GPL must include the source code. Any modifications made to a GPL source code will also be licensed under GPL. This was to ensure that software once "opened" to the community could not be "closed" at a later time.

In 1998 a non-profit institution called Open Source Initiative (OSI) defined the term "open source software" to emphasize a break with the anti-business past associated with GNU to place a new emphasis in the community on the possibilities of extending the free software model to the commercial world. The OSI does not define a specific license as GPL but lays down the pre-requisites of the distribution terms of open source software. It thereby accepts various licenses whose distribution terms comply with the Open Source Definition (OSD). There are ten criteria mentioned at the OSI Web site (http://www.opensource.org/docs/osd). The main intentions for the OSD are described in the article "*Open source licensing, Part 1: The intent*" [1] by Martin Streicher and are summarized in *Table 3.1* below.

	Intention	**Explanation**
1	Licensees are free to use open source software for any purpose whatsoever.	This basically means that the licensee need not justify the usage of the open source software.
2	Licensees are free to make copies of open source software and are free to distribute those copies without payment of royalties to a licensor.	This implies that the licensee can redistribute the software at a cost or free of charge.
3	Licensees are free to create derivative works of open source software and are free to distribute those works without payment of royalties to a licensor.	This allows the licensee to modify the open source software and then redistribute it with or without charge. The licensee is in no way liable to pay any amount to the licensor for this neither can the licensor pose any restrictions on the derivative works. A pre-requisite for intention 3 to occur is the pre-existence of intention 4.
4	Licensees are free to access and use the source code of open source software.	This means the source code is freely available
5	Licensees are free to combine open source and other software.	This gives the licensee the ability to mix open source software with other software.

Table 3.1 - Intentions of open source software

We review in the next section how the GPL and MIT licenses meet the above intents of open source software.

3.2.2 Commonly used open source licenses

Though there are over 50 OSI approved licenses most of the licenses fall under two categories:

- **Academic licenses,** such as the Berkeley Software Distribution (BSD) license, allow software to be used for any purpose. Software obtained via an academic license can be freely changed, sold, redistributed, sublicensed, and combined with other software.
- **Reciprocal licenses** like the GNU General Public License (GPL), also allow software to be used for any purpose, however it enforces that the changed or modified software must also be licensed under the exact terms as the original license.

A GPL licensed code does not allow proprietary software to link to it. It also does not permit redistribution with software having a GPL non-compatible license. Also redistribution of the derivative works need to be with GPL. On the other hand MIT licensed software allows all of it. It permits proprietary code to link to it, redistribution with non-MIT license software and redistribution of derivative works with non-MIT license. Interestingly, they both are open source software licenses as they follow the open source definition specified by the OSI. *Table 3.2* compares the GPL versus the MIT license.

	GPL license	**MIT license**
Allow proprietary code to link to open source code	No	Yes
Allow redistribution of software with other code that has another license	No	Yes
Allow redistribution of derivative work	Yes. Derivative work becomes open source with GPL license	Yes

Table 3.2 - Comparing the GPL vs. the MIT reciprocal licenses

Let's take an even closer look at the GPL and MIT licenses by reviewing excerpts of each license as shown in *Listing 3.1* and *3.2* respectively. We will verify each of the licenses satisfy the five intentions explained earlier in *Table 3.1*

You may modify your copy or copies of the Program or any portion of it, thus forming a work based on the Program, and copy and distribute such modifications or work -- provided that you -- cause any work that you distribute or publish, that in whole or in part contains or is derived from the Program or any part thereof, to be licensed as a whole at no charge to all third parties under the terms of this License.

These requirements apply to the modified work as a whole. If identifiable sections of that work are not derived from the Program, and can be reasonably considered independent and separate works in themselves, then this License, and its terms, do not apply to those sections when you distribute them as separate works.

But when you distribute the same sections as part of a whole which is a work based on the Program, the distribution of the whole must be on the terms of this License, whose permissions for other licensees extend to the entire whole, and thus to each and every part regardless of who wrote it.

Listing 3.1 - An excerpt of the GNU license

The sentence *"You may modify your copy or copies of the Program or any portion of it, thus forming a work based on the Program, and copy and distribute such modifications or work"* indicates that licensees are free to access, and use open source software for any purpose and that they are free to create and distribute derivative works (Intention 1, 3, 4 and 5). In the same sentence, the clause *"copy and distribute such modifications or work"* further indicates that licensees are free to make copies of the open source software (Intention 2).

Copyright (c) year, copyright holders
Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

< text that follows has been removed for the sake of brevity >

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

Listing 3.2 - An excerpt of the MIT license

The phrase *"permission is hereby granted, free of charge"* clearly indicates that MIT permits licensees to access and use open source software for any purpose whatsoever (Intention 1). Further the phrase *"rights to use, copy, modify, merge, publish, distribute, sublicense and/or sell copies"* combined with *"free of charge"* indicates that licensees are free to make and distribute copies, access the source code, create derivative works and combine open source and other software without payment of royalties (Intention 2, 3, 4 and 5).

In conclusion, both GPL and MIT are valid open source licenses as they satisfy the same intentions though through different means, and are thereby best suited in different situations.

Did you know?

The IBM Public License (IPL) was IBM's first open source license. The Common Public License (CPL) is essentially the next version of the IPL. In 2009 CPL was superseded by the Eclipse Public License (EPL).

3.3 Choosing the right license

By now you must be feeling more of a lawyer and less of a developer. Yet, all of this information is very relevant to you: It can help you avoid a lawsuit. An interesting licensing

violation is the case between the Free Software Foundation (FSF) and Cisco. In 2004 a German court ordered Cisco to stop selling its wireless routers because it was in violation of the terms and conditions of the GPL license - Cisco's products were using GPL-licensed software in their code, but not providing free availability of their source code.

Another case of license violation happened early in 2009 when Microsoft was found in violation of the GPL license on the Hyper-V code it released to the open source community. More information can be found at this link:

http://www.theregister.co.uk/2009/07/23/microsoft_hyperv_gpl_violation/.

So what is the right open source license to use for a given software? There is no magic formula but some of the things to consider are listed below:

- Is the work in question a derived work? Or would the work be redistributed with other open source software? If so, what are the license terms you have agreed as a licensee?
- Do you want the software to be revenue generating? Review the different open source business models as described earlier in *Chapter 2*.
- Do you want access to use licensees' contributions? How do you add this clause to your license?
- Review the licenses in the OSI website - http://www.opensource.org/licenses to see if any of the existing and approved licenses of OSI would be good for you to use.

This and several other factors are things to consider while choosing an appropriate license for the software in question. Depending on the scope of the project you could get into legal discussions to help you chose an appropriate license.

3.4 Exercises

The OSI categorizes various approved licenses. Review these categories and the licenses' terms and conditions at http://www.opensource.org/licenses/category

3.5 Summary

In this chapter you learned about the concept of Intellectual Property and how it related to a license. You also learned about the relationship between a license, licensor and licensee. Next, the chapter explained what makes a license an open source license, and provided a description and comparison between the BSD, GPL and MIT licenses. Finally the chapter provided questions that would help you choose the most appropriate open source license for your needs.

3.6 Review questions

1. What is required for OSI license approval?
2. Why is it important for developers to know about OSS licenses?

3. Name 3 popular OSS licenses
4. How come the MIT and GPL licenses are both OSS licenses even though they have contradictory items?
5. Name the two main categories of OSS licenses
6. Is the Eclipse Public License approved by OSI?
    A. Yes
    B. No
7. If I modify a program licensed under GPL and distribute the object code of the modified program for free, must I make the source code available?
    A. Yes
    B. No
8. Under MIT, can you link proprietary code to open source code?
    A. Yes
    B. No
9. Intellectual property and copyright are synonyms. True or false?
    A. True
    B. False
10. Which of the following is not an OSS license?
    A. GPL
    B. BSD
    C. MIT
    D. EPL
    E. None of the above